Thalassemia Research: Updated Attitude Scoring Schemas

This document outlines the advanced scoring logic implemented during the attitude calculation refinements. It details the multi-tiered integer assignments and the inverse-frequency probability weighting applied to Partner Selection and Cascade Screening.

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1. The V2 / V3 Penalty Schema

Unlike early scoring phases which simply awarded +1 for any positive attitude, this updated methodology applies a stretched Likert-style scale. It heavily penalizes dangerous medical attitudes (e.g., -3) while rewarding highly protective attitudes (e.g., +2). This shatters statistical “ceiling effects” and creates continuous data distributions.

A. Partner Selection Attitudes

(Note: Q34 was removed in V3 as it represented a barrier/practice rather than a pure attitude).

- **Q28: What should a carrier do after diagnosis?**
  - `Get the partner tested before marriage` **(+2)**
  - `Get family members tested` **(+1)**
  - `I don't know` **(0)**
  - `Ignore it` **(-3)**
- **Q30: Are you willing to / Do you have a consanguineous marriage?**
  - `Definitely not` **(+2)**
  - `Not sure` **(-1)**
  - `Yes I am willing` or `Yes I have` **(-3)**
- **Q31: Do you accept marriage between two thalassemia carriers?**
  - `No` **(+2)**
  - `Not sure` **(-1)**
  - `Yes` **(-3)**
- **Q32: How important is thalassemia screening before marriage?**
  - `Very important` **(+2)**
  - `Important` **(+1)**
  - `Not sure` **(-1)**
  - `Not important` **(-2)**

B. Cascade Screening Attitudes

(Note: Q36 was removed in V3 as it is a practice metric, not an attitude).

- **Q35: Is it important for your family members to undergo screening?**
  - `Agree` **(+2)**

- `Don't know` **(0)**
- `Disagree` **(-2)**

- **Q39: How easy is it to convince relatives to undergo screening?**
  - `Very easy` **(+2)**
  - `Somewhat easy` **(+1)**
  - `Not sure` **(0)**
  - `Difficult` **(-1)**
- **Q40: How important is cascade screening in thalassemia prevention?**
  - `Very important` **(+2)**
  - `Important` **(+1)**
  - `Slightly important` **(-1)**
  - `Not important` **(-2)**

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2. The "Weighted V3" Schema (Cross-Question Inverse Frequency)

To make the data mathematically robust against extremely common baseline responses, the **Weighted V3** schema applies probability calculus on top of the assigned integer weights.

Mathematical Formula:

$$\text{Final Score} = \text{Assigned Weight} \times (1 - p)$$

(Where p is the proportion of the total 201 participants who selected that exact answer).

Psychological Implication:

1. **Common Protective Attitude:** If a participant selects "Definitely not" for consanguineous marriage (+2 weight) and 85% of the cohort also selected it ($p = 0.85$), their reward is scaled down: `+2 * 0.15 = +0.30`. They are rewarded, but only slightly, because this attitude is medically expected.
2. **Rare Dangerous Misconception:** If a participant selects "Yes" to consanguineous marriage (-3 penalty weight) and only 5% of the cohort selected it ($p = 0.05$), their penalty remains massive: `-3 * 0.95 = -2.85`.

By applying this calculus, the algorithm creates a highly sensitive, truly continuous spectrum that perfectly isolates dangerous medical outliers for targeted educational intervention.